Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/12/2024
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Investigating Structure and Function of the Eye		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/30/2019	**End Date:**	09/30/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2033 per HHC element/JSC (Ed., 12/16/21) NOTE: End date changed to 1/24/2026 per HHC element/JSC (Ed., 4/8/21) NOTE: End date changed to 9/30/2025 per PI (Ed., 12/21/19)		
Key Personnel Changes/Previous PI:	Removed SMC Lee; he is now working in the HRP-HHC Element office. Adding Co-I Jennyffer Smith, who is a new member of the CVL and study team.		

COI Name (Institution):	Brunstetter, Tyson O.D., Ph.D. (NASA Johnson Space Center) Huang, Alex M.D., Ph.D. (Doheny Eye Institute) Karanjia, Rustum M.D., Ph.D. (Doheny Eye Institute) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Martin, Bryn Ph.D. (University of Idaho, Moscow) Sadda, Srinivas M.D. (Doheny Eye Institute) Smith, Scott Ph.D. (NASA Johnson Space Center) Zwart, Sara Ph.D. (University of Texas, Galveston) Gibson, Charles O.D. (Coastal Eye Associates) Kramer, Larry M.D. (University of Texas Health Science Center, Houston) Lytle, Jason Ph.D. (KBR/NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Smith , Jennyffer (KBR)
Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This proposal will identify if ocular structure and function alterations occur at a greater frequency and magnitude during one-year missions compared to six-month and two-month expeditions and whether the recovery profile is dependent upon mission duration. In addition, this project will determine if changes in vascular structure and function are greater after one-year missions and if they contribute to alterations in ocular structure and function. The identification of structural and functional changes related to the development, progression, and recovery of Spaceflight Associated Neuro-ocular Syndrome (SANS) will provide NASA the information necessary to inform the risk posture for future interplanetary expeditions with duration of up to three years and to identify possible countermeasures. SPECIFIC AIMS: Conducting SANS research on two-month, six-month, and one-year crewmembers will enable us to objectively generate data to help NASA determine if and how SANS findings change with mission duration. Data from these missions will enable the development of a non-linear trend model that can be extrapolated to make predictions for spaceflights that are up to three years in duration. Therefore, the results of the "Investigating Structure and Function of the Eye" (iSAFE) study will help NASA to define the risk posture for future interplanetary expeditions and to identify possible countermeasures. Specific Aim 1: To determine if ocular structural changes develop to a greater degree (frequency or magnitude) during long-duration one-year spaceflight missions compared to findings during shorter length missions, and if recovery is prolonged after longer missions. Specific Aim 2: To determine if ocular vascular structure is altered to a greater degree during long-duration one-year spaceflight missions, and if recovery is more prolonged. Specific Aim 3: To determine if ocular function is altered to a greater degree during long duration one-year spaceflight missions, and the recovery profile. Specific Aim 4: To determine if measures of vascular structure and function are altered to a greater degree during long-duration one-year spaceflight missions, and if these vascular adaptations correlate with alterations in ocular structure and function. RELEVANCE AND MAP TO THE NASA HUMAN RESEARCH ROADMAP: This multi-project proposal is in response to NASA research announcement Human Exploration Research Opportunities (HERO), 80JSC017N0001-BPBA, Appendix C, Topic 1: Analyses of the Temporal Nature of Human Adaptation to Long-Duration Low-Earth Orbit Mission Virtual NASA Specialized Center of Research (VNSCOR). This proposal addresses multiple Human Research Program (HRP) Integrated Research Plan Gaps, including: • SANS101: Determine the relationship between fluid shifts (intravascular, interstitial, CSF) and ocular manifestations in astronauts during spaceflight. • SANS102: Determine the relationship between the fluid-shifts induced ocular changes and fluid shifts in the CNS, including whether elevated intracranial pressure or brain edema play a role.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Investigating Structure and Function of the Eye (iSAFE) research study will advance NASA's understanding of Spaceflight Associated Neuro-ocular Syndrome (SANS), an important human health and performance risk. This goal will be accomplished by quantifying how ocular alterations develop, progress, and recover as a function of spaceflight duration and by identifying underlying mechanisms. Results from this study are anticipated to lead to a temporal model of SANS progression during long-term missions and to inform the development of countermeasures. Given the unique environment of the International Space Station, commonly used ophthalmic instruments are being implemented in novel operational environments (e.g., electroretinography (ERG), optical coherence tomography (OCT) angiography, pneumotonometry). This work may lead to the adoption of these new hardware, software, or protocol elements in clinical practice, benefiting patients on Earth. SANS shares characteristics with several terrestrial ophthalmic diseases, such as papilledema, and iSAFE study results could provide new insights into mechanisms underlying these conditions.
	INTRODUCTION Spaceflight-associated neuro-ocular syndrome (SANS) describes structural and functional alterations of the eye that develop in many astronauts during long-duration spaceflight missions (>1 month). The chronic headward fluid shift that occurs in weightlessness is hypothesized to be the primary factor underlying this condition. However, the specific mechanisms are unknown, and it remains uncertain whether the incidence and severity of SANS findings correlate with mission duration. PURPOSE The purpose of this study is to evaluate whether spaceflight mission duration influences SANS incidence and severity and whether vascular changes during weightlessness relate to SANS findings. Specifically, we will test the hypotheses that ocular tissue structure, ocular vascular structure, and ocular function are altered in spaceflight. Cervicothoracic vasculature will be evaluated for volume increase in both incidence and magnitude with mission duration, as well as the magnitude of vascular adaptation and its correlation with ocular findings. Moreover, we will test the hypothesis that a genetic predisposition to SANS is conferred by 1-carbon metabolism pathway-associated gene variants that compromise vascular structure and function by perturbing of B-vitamin and related metabolite levels. METHODS Advanced techniques will be used to study the eyes, cervicothoracic vasculature, and 1-carbon pathway genetics and biochemistry in 30 astronauts scheduled to participate in missions on the International Space Station (ISS).

Task Progress:	Subjects will be divided into 3 cohorts (n=10/cohort) based on mission duration, specified as Short (~2 months), Standard (~6 months), or Extended (~1 year). Ocular structure and function will be evaluated using magnetic resonance imaging (MRI) and 3D structural analysis, optical coherence tomography (OCT), OCT angiography (OCTA), electroretinography (ERG), visual field analysis, pneumotonometry (PTM, intraocular pressure/IOP), visual field analysis, optical biometry, and dynamic vessel analysis (DVA). Ultrasound will be used to investigate variables associated with the cervicothoracic vasculature, including stroke volume, cardiac output, and internal jugular vein morphology/pressure and carotid intima-media thickness. In addition, MRI will be used to assess brain structural changes. Many of the experiments will be performed before, during, and after flight (Table 1). Some of the data collected on the ground (i.e., before and after flight) will be acquired in the seated and supine postures to simulate the fluid shift experienced upon transition from 1 g to weightlessness. To complement the research measures, medically relevant test results already collected by the Space Medicine Operations Division will be requested. Mission Impact The Investigating Structure and Function of the Eye (iSAFE) research study will advance NASA's understanding of SANS, an important human health and performance risk through the following specific aims: 1) The relationships between mission duration and SANS development, progression, and recovery will be defined. 2) Experiments will identify whether optic disc edema is accompanied by leakage of the retinal vasculature. 3) With the implementation of ERG, visual function will be measured during spaceflight for the first time and will be related to the development of optic disc edema. 4) Possible mechanisms underlying SANS will be evaluated, including vascular adaptations in weightlessness and 1-carbon pathway genetics. This research will narrow the SANS101 and SANS102 knowledge gaps described in the Human Research Roadmap and, in turn, will help NASA to evaluate this spaceflight-associated risk more accurately and refine its approach to SANS countermeasure development. Study Progress in this reporting period includes a total of 8 of 10 crew assigned to standard duration missions (4-6 months) who have consented to participate. Forward work includes continuing to support all phases of preflight, inflight, and postflight data collection and training of crew members.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Macias BR, Laurie SS, Brunstetter T, Young M, Huang A, Karanjia R, Lee SMC, Martin BA, Kramer LA, Sadda S, Smith SM, Zwart SR, Martin DS, Miller A, Melvin SA. "CIPHER: Investigating structure and function of the eye (isafe)." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024